

A Probabilistic Shared Control Framework for SE(3) Grasping

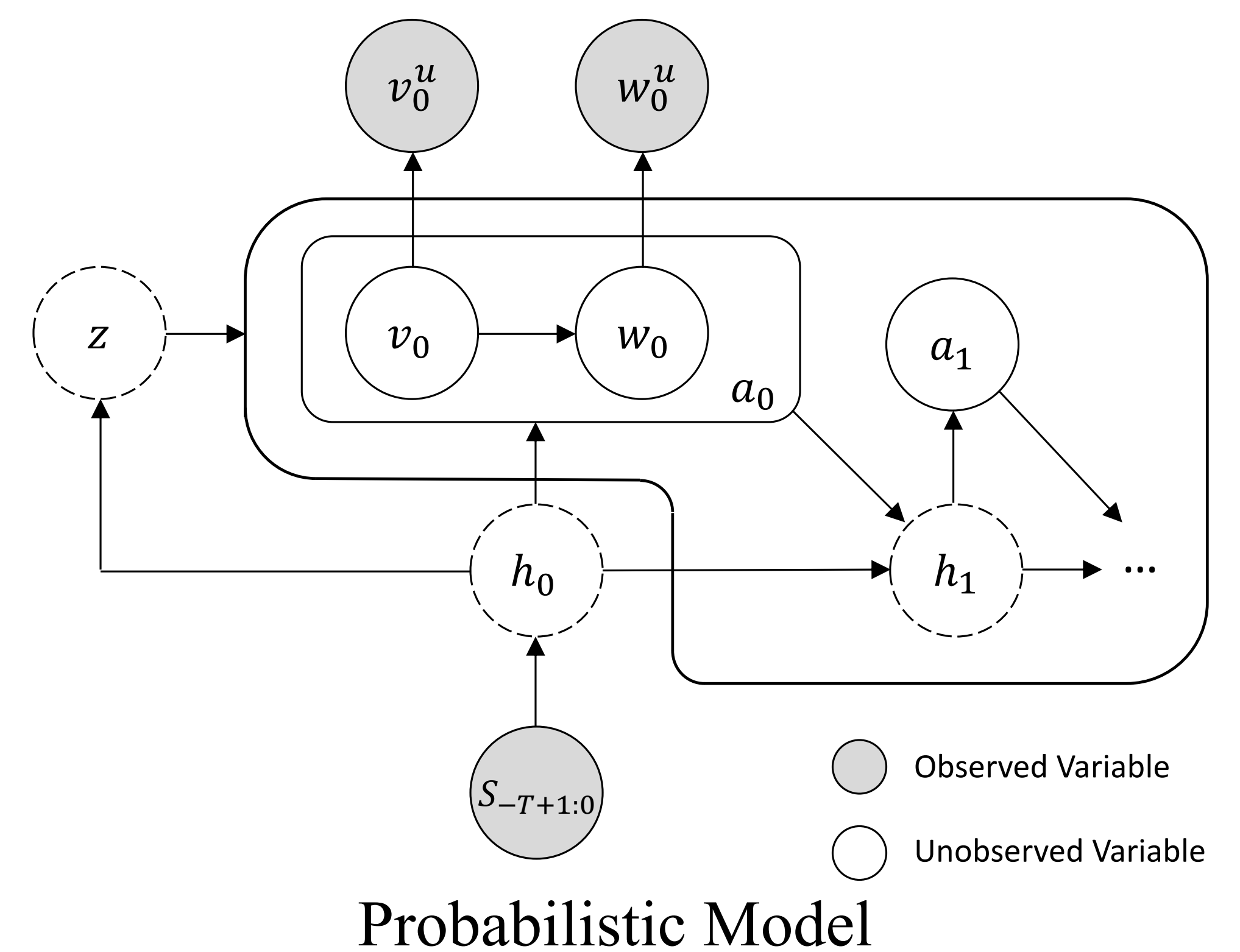
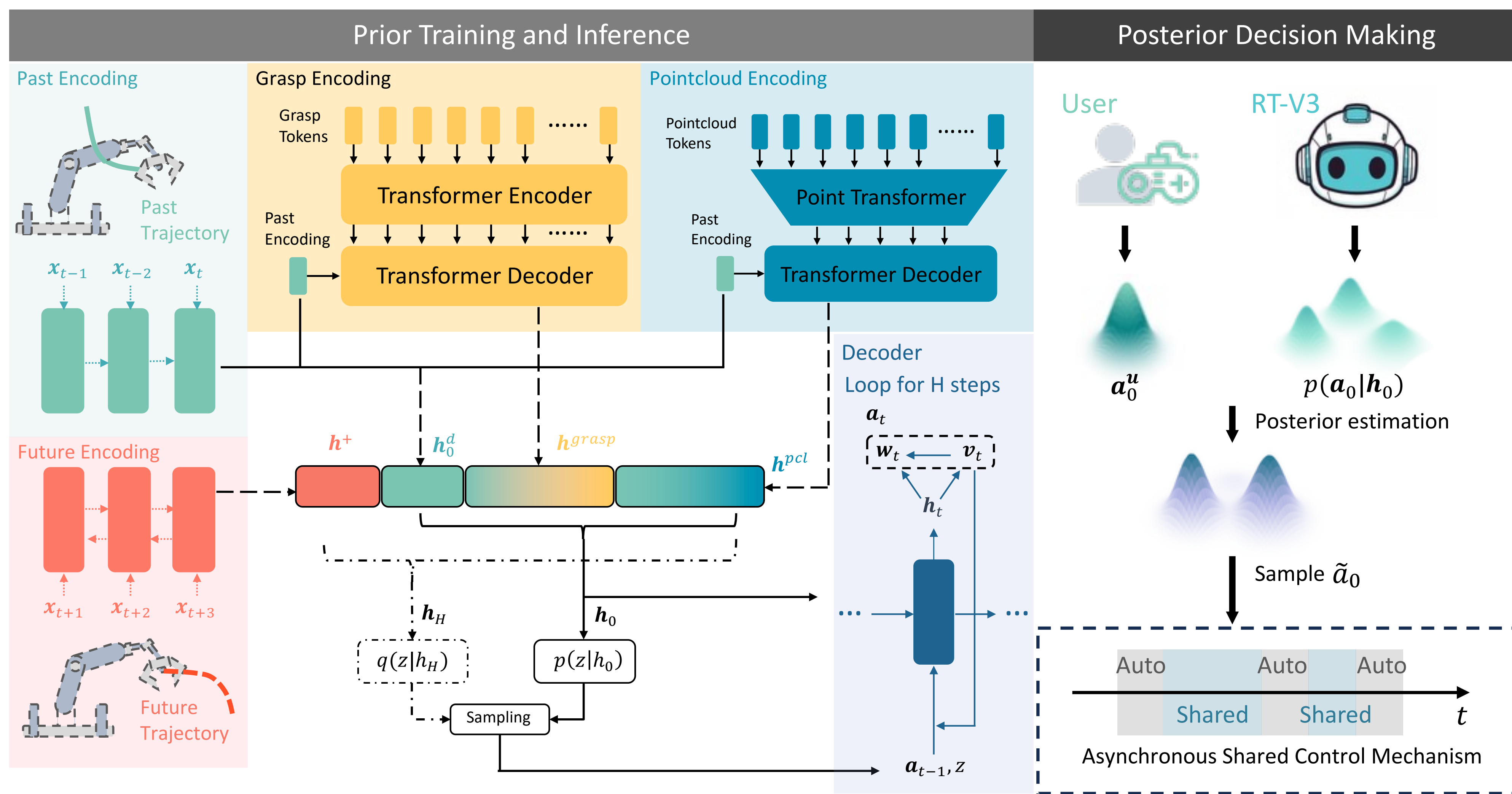
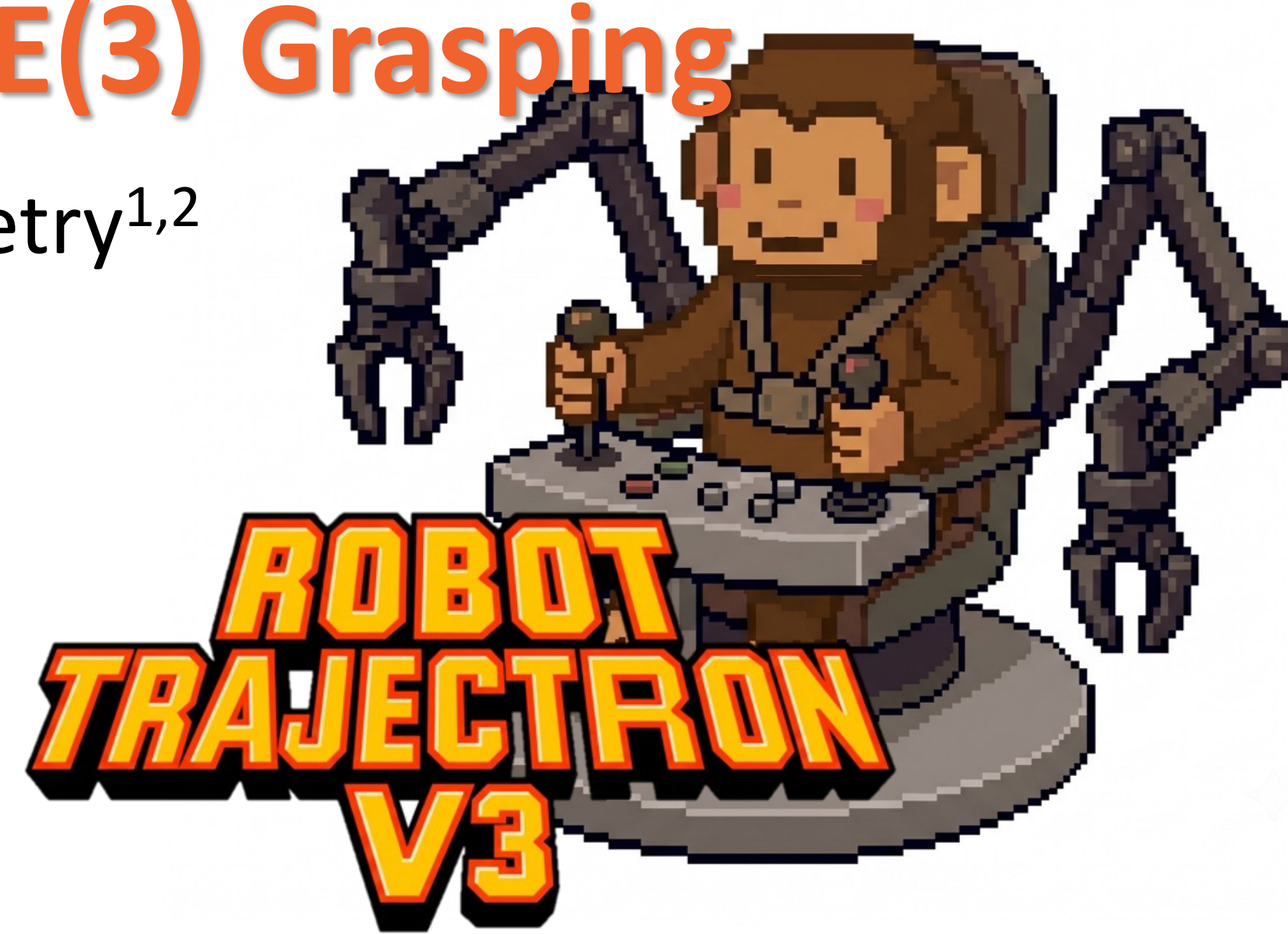
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Flanders Make²

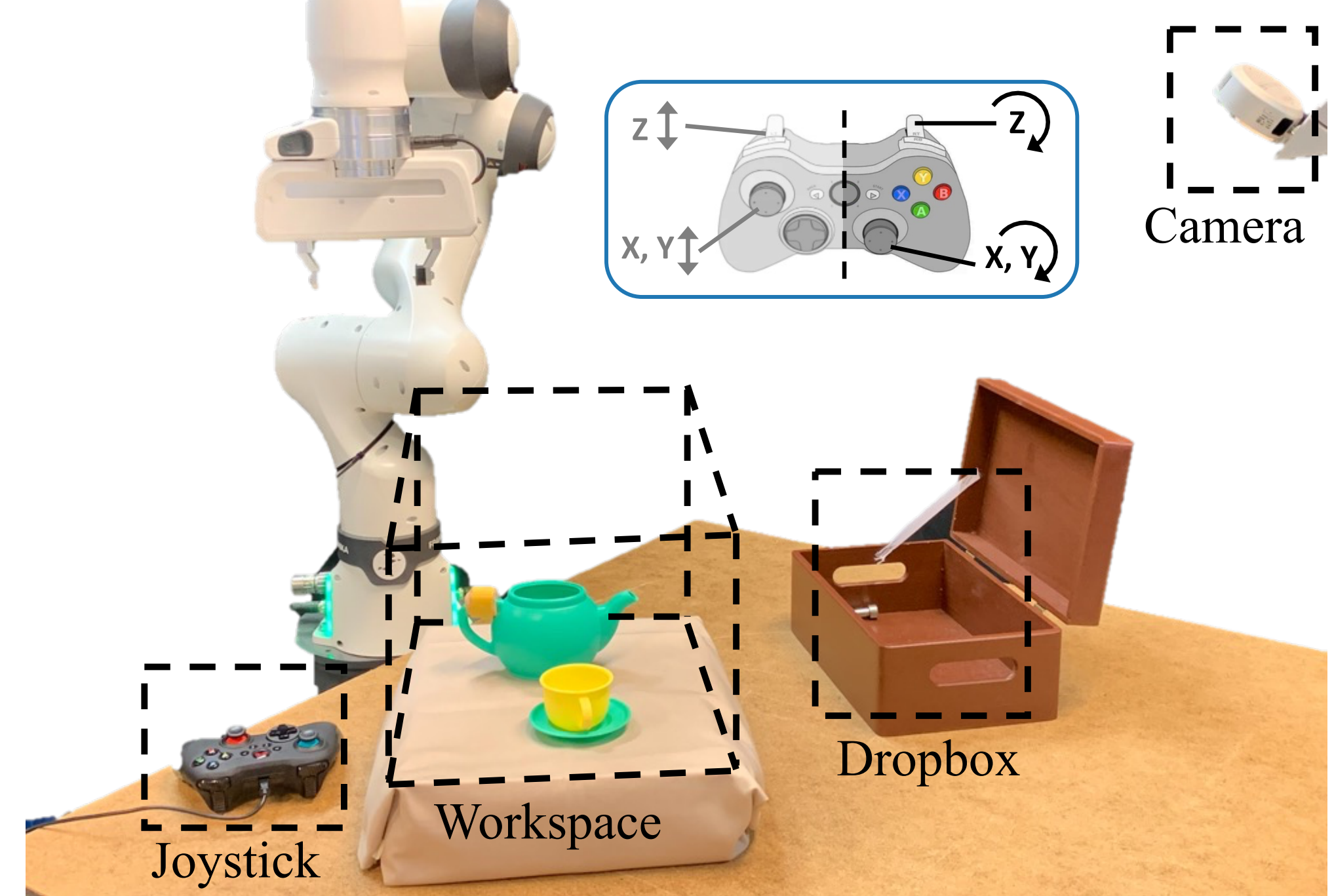


Motivation: Assist users with low-bandwidth, noisy interfaces in controlling high-DoF robots for grasping tasks.

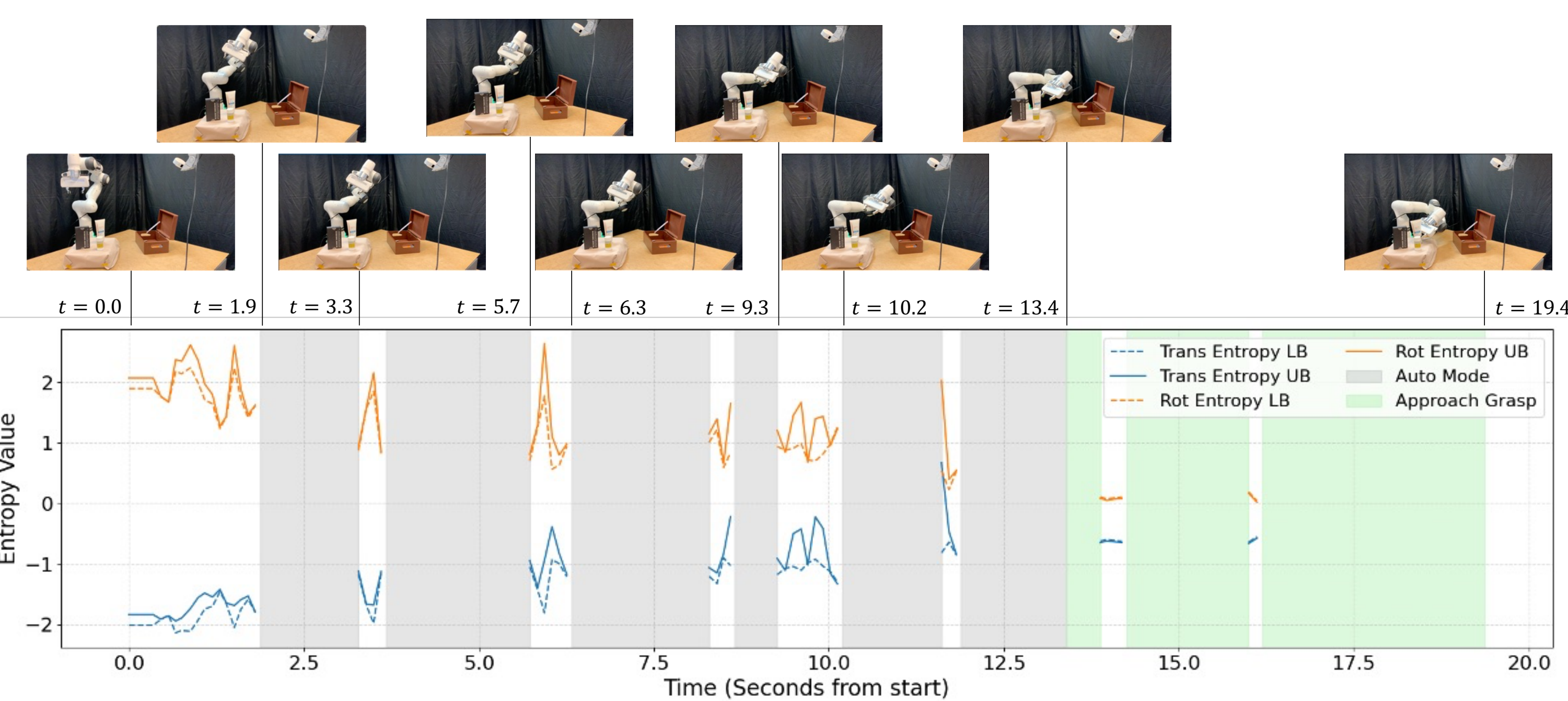
Challenge: Infer the intended grasp affordance instead of the intended object

Solution:

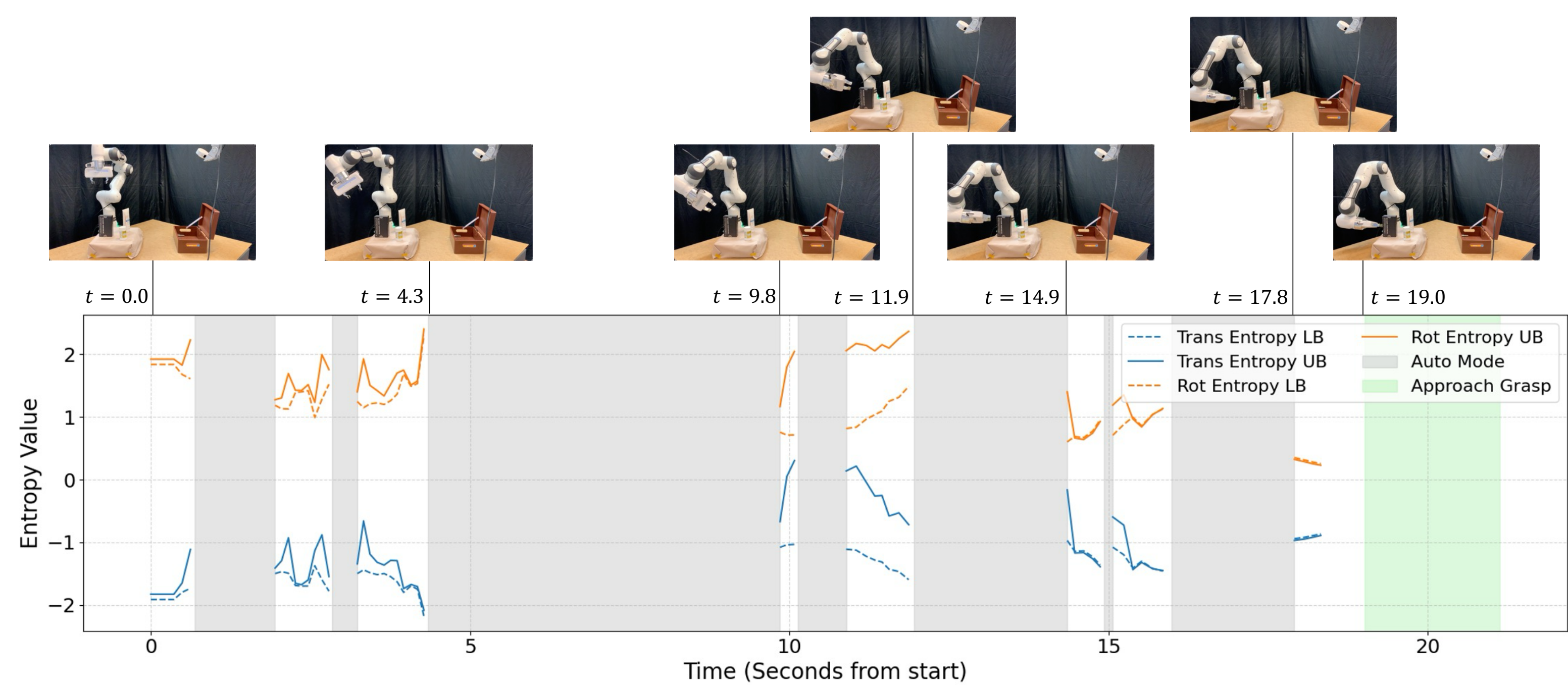
- A trajectory-level probabilistic shared-control framework for multi-object, multi-affordance grasping.
- Bayesian inference that combines scene-conditioned intent priors with real-time user commands.



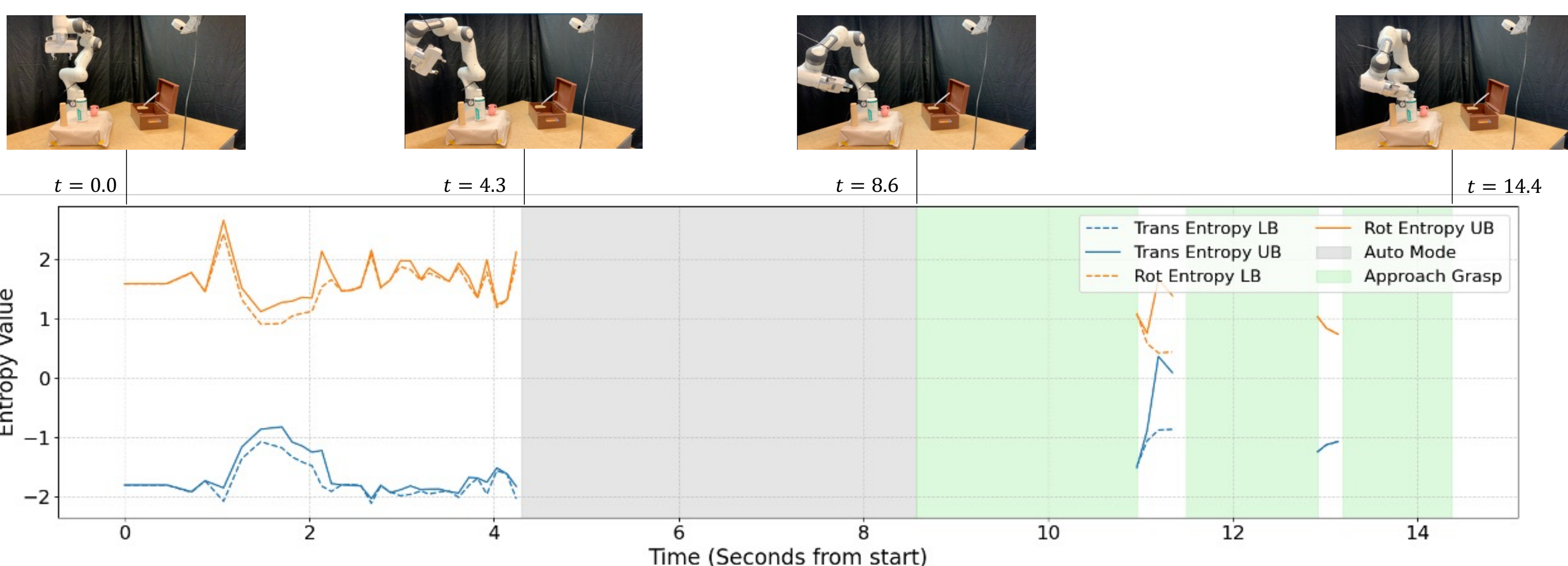
Experimental Setup



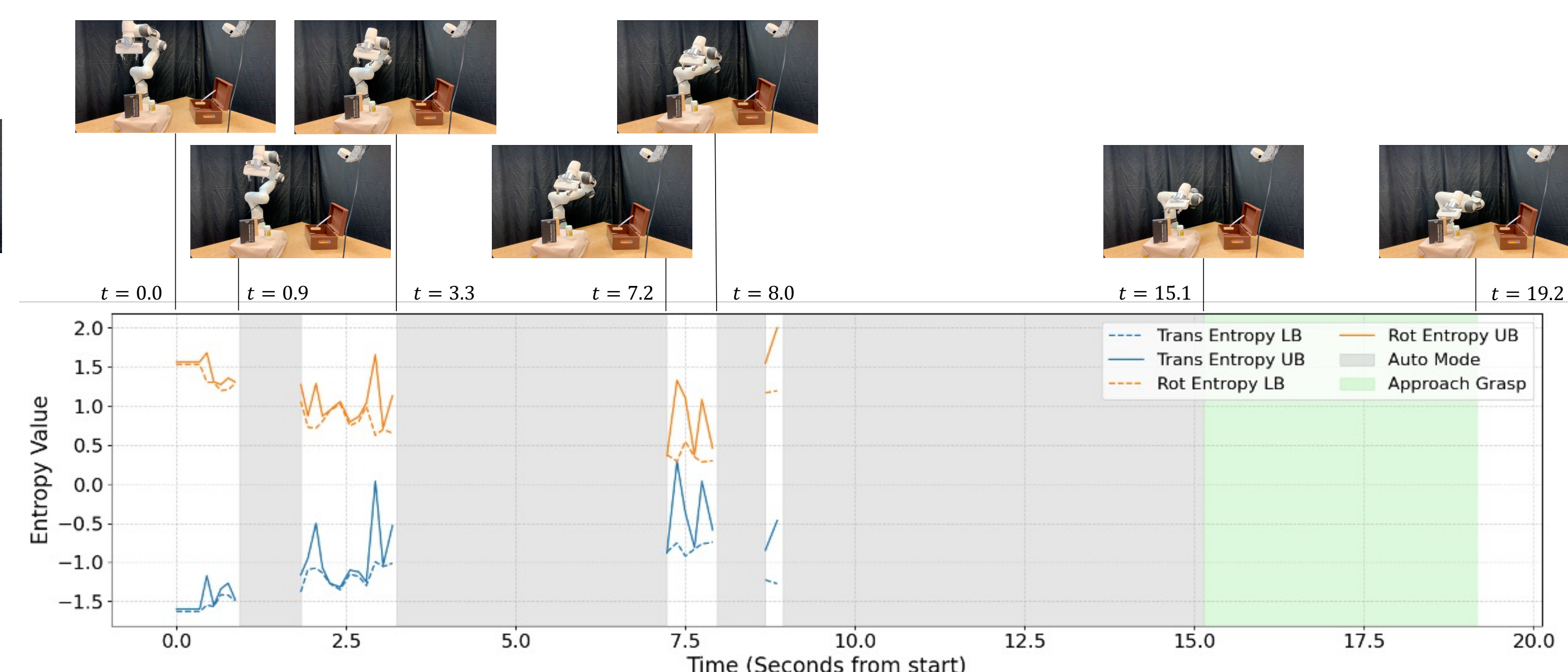
Use Case 1: Grasp the facial cleanser from the right.



Use Case 2: Grasp the box from the left.



Use Case 3: Grasp the cap of the bottle from the left.



Use Case 3: Grasp the sunscreen from the back side.

- Song, P., Li, Z., Fu, Z., Rios, F.U. and Detry, R., 2026. Robot Trajectron V3: A Probabilistic Shared Control Framework for SE (3) Manipulation. *arXiv preprint arXiv:2607.09315*.
- Song, P., Du, Y., Saussus, O., De Schrijver, S., Caprara, I., Janssen, P. and Detry, R., 2025. Robot trajectron V2: A probabilistic shared control framework for navigation. *IJRR 2026*.